

- 1 CM3
- 2 Overview of machines, contexts and proof obligations
- 3 Proof Obligations for Contexts and Machines
 - PO thm/THM (context)
 - PO th/THM (machine)
 - PO evt/inv/INV
 - PO evt/act/FIS
- 4 Method for proving invariance properties

- 5 Verification
 - Definition of machine for checking contract
 - Soundness of the technique
- 6 Conclusion

Current Section Summary

① CM3

② Overview of machines, contexts and proof obligations

③ Proof Obligations for Contexts and Machines

PO thm/THM (context)

PO th/THM (machine)

PO evt/inv/INV

PO evt/act/FIS

④ Method for proving invariance properties

⑤ Verification

Definition of machine for checking contract

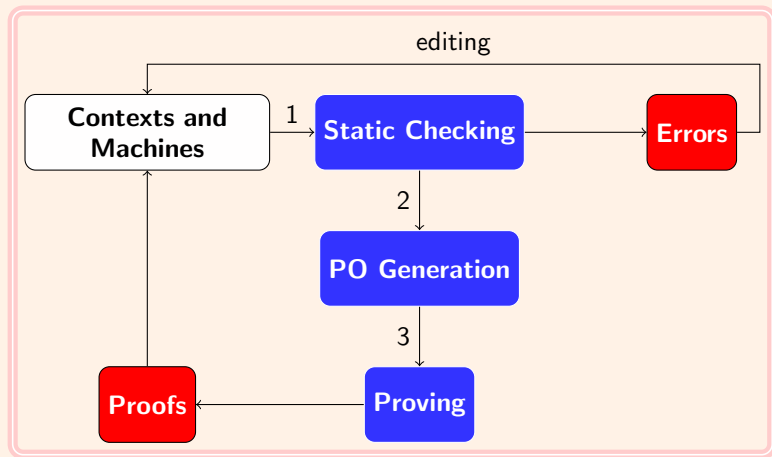
Soundness of the technique

⑥ Conclusion

Current Section Summary

- ① CM3
- ② Overview of machines, contexts and proof obligations
- ③ Proof Obligations for Contexts and Machines
 - PO thm/THM (context)
 - PO th/THM (machine)
 - PO evt/inv/INV
 - PO evt/act/FIS
- ④ Method for proving invariance properties
- ⑤ Verification
 - Definition of machine for checking contract
 - Soundness of the technique
- ⑥ Conclusion

Analysis of the Event-B Models



MACHINE

m

REFINES

am

SEES

c

VARIABLES

u

INVARIANTS

$I(s, c, u)$

THEOREMS

$Q(s, c, u)$

VARIANT

$exp(s, c, u)$

EVENTS

INITIALIZATION

...

e

...

END

MACHINE

m

REFINES

am

SEES

c

VARIABLES

u

INVARIANTS

$I(s, c, u)$

THEOREMS

$Q(s, c, u)$

VARIANT

$exp(s, c, u)$

EVENTS

INITIALIZATION

...

e

...

END

- $\Gamma(m)$: environment for the machine m defined by the context c and it provides a list of seen axioms $Ax(s, c)$ and a list of seen theorems $Th(s, c)$ for the sets s and constants c .
- $\Gamma(m) \vdash \forall u. \text{INIT}(s, c, u) \Rightarrow I(s, c, u)$
- For each event e in E :
 $\Gamma(m) \vdash \forall u, u'. I(s, c, u) \wedge BA(e)(u, u') \Rightarrow I(u')$
- For each event e in E :
 $\Gamma(m) \vdash \forall u. I(s, c, u) \wedge GRD(e)(s, c, u) \Rightarrow \exists u'. BA(e)(u, u')$
- $\Gamma(m) \vdash \forall u. I(s, c, u) \Rightarrow Q(s, c, u)$
- Generated proof obligations are derived from those conditions.

Checking the well formation of Event-B expressions

- Event-B expressions are contexts, machines, properties, equations, set-theoretical expressions ...
- e is an Event-B expression and $\text{wd}(e)$ is a logical property expressing the well definition of e .
- $\text{wd}(1 = 2) \triangleq \text{wd}(1) \wedge \text{wd}(2)$
- $\text{wd}(a/b) \triangleq b \neq 0 \wedge \text{wd}(a) \wedge \text{wd}(b)$
- $\text{wd}(f(g)) \triangleq g \in \text{dom}(f) \wedge f \in A \leftrightarrow B$

Current Section Summary

- ① CM3
- ② Overview of machines, contexts and proof obligations
- ③ Proof Obligations for Contexts and Machines
 - PO thm/THM (context)
 - PO th/THM (machine)
 - PO evt/inv/INV
 - PO evt/act/FIS
- ④ Method for proving invariance properties
- ⑤ Verification
 - Definition of machine for checking contract
 - Soundness of the technique
- ⑥ Conclusion

- ① CM3
- ② Overview of machines, contexts and proof obligations
- ③ Proof Obligations for Contexts and Machines
 - PO thm/THM (context)
 - PO th/THM (machine)
 - PO evt/inv/INV
 - PO evt/act/FIS
- ④ Method for proving invariance properties
- ⑤ Verification
- ⑥ Conclusion

CONTEXTS

c

EXTENDS

ac

SETS

s

CONSTANTS

c

AXIOMS

$Ax(s, c)$

THEOREMS

$th_1 : P_1(s, c)$

...

$th_n : P_n(s, c)$

$th : P(s, c)$

...

END

s

seen sets

c

seen constants

$Ax(s, c)$

seen axioms

$Th(s, c)$

previous proved theorems

$PTh(s, c) = \{P_i(s, c) | i 1..n\}$

$P(s, c)$

property over s and c

PO th/THM

$Ax(s, c), Th(s, c) \vdash P(s, c)$

- ① CM3
- ② Overview of machines, contexts and proof obligations
- ③ Proof Obligations for Contexts and Machines
 - PO thm/THM (context)
 - PO th/THM (machine)
 - PO evt/inv/INV
 - PO evt/act/FIS
- ④ Method for proving invariance properties
- ⑤ Verification
- ⑥ Conclusion

post

PO th/THM (machine)

MACHINE

m

...

VARIABLES

u

INVARIANTS

$I(s, c, u)$

THEOREMS

$Q(s, c, u)$

$th : P(s, c, u)$

...

END

s

seen sets

c

seen constants

u

variables

$Ax(s, c)$

seen axioms

$Th(s, c)$

seen theorems

$I(s, c, u)$

invariants

$Q(s, c, u)$

theorems

$P(s, c, u)$

property over s, c and u

PO th/THM

$Ax(s, c), Th(s, c), I(s, c, u) \vdash P(s, c, u)$

- ① CM3
- ② Overview of machines, contexts and proof obligations
- ③ Proof Obligations for Contexts and Machines
 - PO thm/THM (context)
 - PO th/THM (machine)
 - PO evt/inv/INV
 - PO evt/act/FIS
- ④ Method for proving invariance properties
- ⑤ Verification
- ⑥ Conclusion

PO evt/inv/INV

EVENT evt

ANY x **WHERE**

$G(x, s, c, u)$

THEN

$u : \mid BAP(x, s, c, u, u')$

END

$BA(\text{evt}) \hat{=}$

$\exists x. \left(\begin{array}{l} \wedge G(x, s, c, u) \\ \wedge BAP(x, s, c, u, u') \end{array} \right)$

$GRD(\text{evt}) \hat{=} G(x, s, c, u)$

$ACT(\text{evt}) \hat{=} BAP(x, s, c, u, u')$

s

c

u

$Ax(s, c)$

$Th(s, c)$

$I(s, c, u)$

$Q(s, c, u)$

evt

x

$G(x, s, c, u)$

$BAP(x, s, c, u, u')$

$inv : inv(s, c, u')$

seen sets

seen constants

variables

seen axioms

seen theorems

invariants

theorems

event name

event parameter

event guard

event before-after predicate

specific modified invariant

PO evt/inv/INV

$Ax(s, c), Th(s, c), I(s, c, u), G(x, s, c, u), BAP(x, s, c, u, u') \vdash inv(s, c, u')$

PO Q/THM $Ax(s, c), Th(s, c), I(s, c, u) \vdash Q(s, c, u)$

- ① CM3
- ② Overview of machines, contexts and proof obligations
- ③ Proof Obligations for Contexts and Machines
 - PO thm/THM (context)
 - PO th/THM (machine)
 - PO evt/inv/INV
 - PO evt/act/FIS
- ④ Method for proving invariance properties
- ⑤ Verification
- ⑥ Conclusion

EVENT evt

ANY x **WHERE**

$G(x, s, c, u)$

THEN

$u : |BAP(x, s, c, u, u')$

END

$BA(\text{evt}) \hat{=}$

$\left(\begin{array}{l} \wedge G(x, s, c, u) \\ \wedge BAP(x, s, c, u, u') \end{array} \right)$

$GRD(\text{evt}) \hat{=} G(x, s, c, u)$

$ACT(\text{evt}) \hat{=}$

$BAP(x, s, c, u, u')$

s

c

u

$Ax(s, c)$

$Th(s, c)$

$I(s, c, u)$

$Q(s, c, u)$

evt

x

$G(x, s, c, u)$

$BAP(x, s, c, u, u')$

seen sets

seen constants

variables

seen axioms

seen theorems

invariants

theorems

event name

event parameter

event guard

event before-after predicate

PO evt/act/FIS

$Ax(s, c), Th(s, c), I(s, c, u), G(x, s, c, u), \vdash \exists u'. BAP(x, s, c, u, u')$

Current Section Summary

- ① CM3
- ② Overview of machines, contexts and proof obligations
- ③ Proof Obligations for Contexts and Machines
 - PO thm/THM (context)
 - PO th/THM (machine)
 - PO evt/inv/INV
 - PO evt/act/FIS
- ④ Method for proving invariance properties
- ⑤ Verification
 - Definition of machine for checking contract
 - Soundness of the technique
- ⑥ Conclusion

Contract verification : what the technique involves

A programme P fulfils a contract $(pre, post)$:

- P transforms a variable x starting from an initial value x_0 and producing a final value x_f : $x_0 \xrightarrow{P} x_f$
- x_0 satisfies pre : $pre(x_0)$ and x_f satisfies $post$: $post(x_0, x_f)$
- $pre(x_0) \wedge x_0 \xrightarrow{P} x_f \Rightarrow post(x_0, x_f)$

```
requires  $pre(x_0)$   
< ensures  $post(x_0, x_f)$   
variables  $X$   
[ begin  
  0 :  $P_0(x_0, x)$   
  instruction0  
  ...  
  i :  $P_i(x_0, x)$   
  ...  
  instructionf-1  
  f :  $P_f(x_0, x)$   
end
```

- $pre(x_0) \wedge x = x_0 \Rightarrow P_0(x_0, x)$
- $pre(x_0) \wedge P_f(x_0, x) \Rightarrow post(x_0, x)$
- verification conditions for all pairs $\ell \longrightarrow \ell'$

Design of a method for proving invariance and safety properties for a programming language

- We consider a classical programming language denoted `PROGRAMS`
- and we assume that this programming language supports assignment, conditional statements, bounded iteration, unbounded iteration, simple or structured variables such as arrays, and the definition of constants.
- Let us consider a programme `P` in `PROGRAMS`; this programme comprises
 - ▶ globally named variables v ,
 - ▶ globally named constants pc ,
 - ▶ types associated with the globally named variables `VALUES` and identified with a set of possible values for the variables,
 - ▶ instructions following an order defined by the syntax of the programming language.

Hypotheses

- We define a set of control points `LOCATIONS`
- for each programme or algorithm `P`, `LOCATIONS` is a finite set of values and a hidden variable denoted `pc` traverses this set according to the sequence.
- the space of possible values `VALUES` is a Cartesian product of the form `LOCATIONS` $\times$ `MEMORY`
- the variables x of the system are decomposed into two independent entities $x = (pc, v)$ subject to the conditions $pc \in \text{LOCATIONS}$ and $v \in \text{MEMORY}$.

$$x = (pc, v) \wedge pc \in \text{LOCATIONS} \wedge v \in \text{MEMORY} \quad (1)$$

We consider an annotated programme P ; we are given a relational model

$$\mathcal{MP} = (Th(s, c), x, \text{VALUES}, \text{INIT}(x), \{r_0, \dots, r_n\}) \text{ where}$$

- $Th(s, c)$ is a theory defining the sets, constants and static properties of this programme
- x is a list of flexible variables, and x comprises a control part and a memory part.
- $LOCATIONS \times MEMORY$ is a set of possible values for x .
- $\{r_0, \dots, r_n\}$ is a finite set of relations linking the values before x and the values after x' , and conforming to the succession relation $\longrightarrow$ between control points.
- $INIT(x)$ defines the set of initial values of (pc_0, v) and $x = (pc_0, v)$ with $pre(v)$ which characterises the initial values of v at the initial point.

Assumptions regarding the calculation

We assume that there exists a graph over the set of control values defining the flow relation, and we denote this structure $(\text{LOCATIONS}, \longrightarrow)$.

Definition

$$\ell_1 \longrightarrow \ell_2 \triangleq pc = \ell_1 \wedge pc' = \ell_2$$

Definition

(Annotation of a control point)

Let $(\text{LOCATIONS}, \longrightarrow)$ be a structure and $\ell \in \text{LOCATIONS}$ a label ℓ . An annotation of a control point ℓ is a predicate $P_\ell(v)$ (assertion) or $P_\ell(v_0, v)$ (relational).



$P_\ell(v_0, v)$ expresses a relation between the initial value of V , denoted v_0 , and v , the current value of V at point ℓ , and therefore $P_\ell(v_0, v) \Rightarrow pre(v_0)$ specifies that v_0 is an initial value.

Relationship between annotation and invariant

- The labels ℓ belong to **LOCATIONS** : $\ell \in \text{LOCATIONS}$.
- The variables v belong to **MEMORY** : $v \in \text{MEMORY}$.
- $pre(v_0)$ specifies the initial values of v .
- Whenever control is at ℓ , v satisfies $P_\ell(v) : pc = \ell \Rightarrow P_\ell(v_0, v)$.
- For any state (ℓ, v) of the programme, the following property is true but must be proved :

$$J(\ell_0, v_0, pc, v) \triangleq \left[\begin{array}{l} \wedge pc \in \text{LOCATIONS} \\ \wedge v \in \text{MEMORY} \\ \dots \\ \wedge pc = \ell \Rightarrow P_\ell(v_0, v) \\ \dots \end{array} \right]$$

- $J(\ell_0, v_0, pc, v)$ is an invariant constructed from the annotations generated, but we must show that this invariant enables us to verify the three conditions of the induction principle.



ℓ_0 denotes the label marking the start of the algorithm and ℓ_f is the end of the programme. We may simply use 0 and f.

Checking to do

$$x = (pc, v) \text{ et } J(\ell_0, v_0, pc, v) \triangleq \left[\begin{array}{l} \wedge pc \in \text{LOCATIONS} \\ \wedge v \in \text{MEMORY} \\ \dots \\ \wedge pc = \ell \Rightarrow P_\ell(v_0, v) \\ \dots \end{array} \right]$$

Let $(Th(s, c), x, \text{VALUES}, \text{INIT}(x), \{r_0, \dots, r_n\})$ a relational model for a program. A property $A(x_0, x)$ is a safety property for P if

$\forall x_0, x \in \text{LOCATIONS} \times \text{MEMORY}. \text{Init}(x_0) \wedge x_0 \xrightarrow{\text{NEXT}} x \Rightarrow A(x)$. We

know that this property implies that there exists a state property $I(x_0, x)$ such that the three properties hold, but we apply this verification to J :

$\forall x_0, x, x' \in \text{LOCATIONS} \times \text{MEMORY} :$

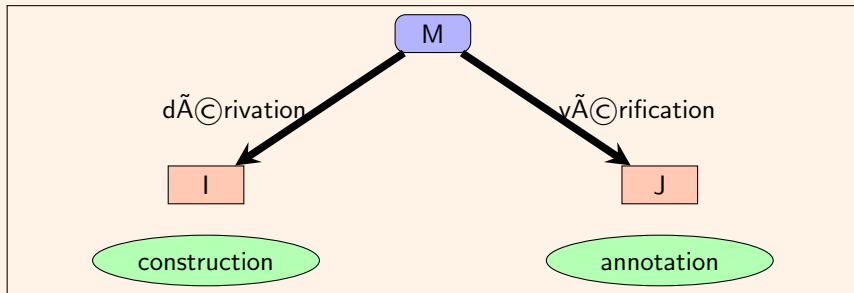
$$\left\{ \begin{array}{l} (1) \text{ INIT}(x_0) \Rightarrow J(x_0, x_0) \\ (2) J(x_0, x) \Rightarrow A(x_0, x) \\ (3) \forall i \in \{0, \dots, n\} : J(x_0, x) \wedge x r_i x' \Rightarrow J(x_0, x') \end{array} \right.$$



$\forall i \in \{0, \dots, n\} : J(x_0, x) \wedge x r_i x' \Rightarrow J(x_0, x')$ is équivalent
to $J(x_0, x) \wedge (\exists i \in \{0, \dots, n\} : x r_i x') \Rightarrow J(x_0, x')$

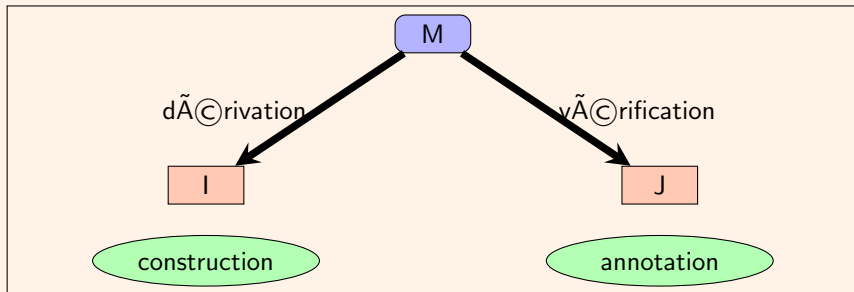
Checking to do

- Application of the correctness to the relational principle of induction : if we verify the three properties, then A is a safety property for the model in question (verification).
- If we wish to show that A is a safety property, then we must use the invariant to construct annotations for the model (derivation).



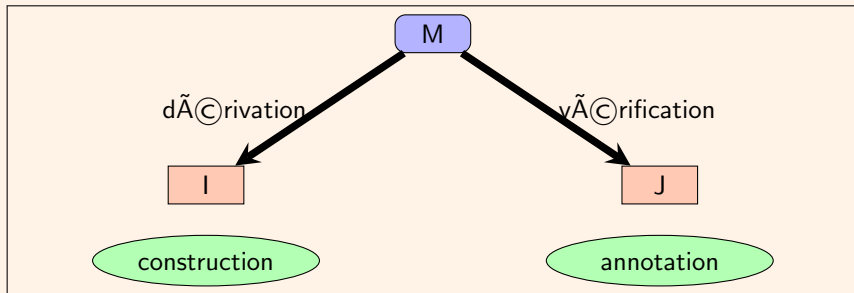
Using the relational induction principle (RI)

- $\text{VALUES} = \text{LOCATIONS} \times \text{MEMORY}$
- $J(pc_0, v_0, pc, v) \triangleq \exists x_0, x \in \text{VALUES}. I(x_0, x) \wedge x = (pc, v) \wedge x_0 = (pc_0, v_0)$ (deduction)
- $I(x_0, x) \triangleq \exists pc_0, pc \in \text{LOCATIONS}, v_0, v \in \text{MEMORY}. J(pc_0, v_0, pc, v) \wedge x = (pc, v) \wedge x_0 = (pc_0, v_0)$ (induction)



Using the assertional induction principle (AI)

- $\text{VALUES} = \text{LOCATIONS} \times \text{MEMORY}$
- $J(pc, v) \triangleq \exists x \in \text{VALUES}. I(x) \wedge x = (pc, v)$ (deduction)
- $I(x) \triangleq \exists pc \in \text{LOCATIONS}, v \in \text{MEMORY}. J(pc, v) \wedge x = (pc, v)$ (induction)



Alignment with programs

- (1) $\forall x_0, x \in \text{VALUES}. \text{Init}(x_0) \wedge \text{NEXT}^*(x_0, x) \Rightarrow A(x) \ (I(x))$
- (2) $\forall x_0, x \in \text{VALUES}. \text{Init}(x_0) \wedge \text{NEXT}^*(x_0, x) \Rightarrow R(x_0, x) \ (IR(x))$

Relations and definitions

$x = (\ell, v)$, $x_0 = (\ell_0, v_0)$, $I(x)$, $IR(x_0, x)$ and annotations $P_\ell(v)$, $RP_\ell(v_0, v)$ are linked in this way :

- $I(x) \triangleq \exists \ell, v. (\ell \in \text{LOCATIONS} \wedge v \in \text{MEMORY} \wedge x = (\ell, v) \wedge P_\ell(v))$
- $P_\ell(v) \triangleq \exists x. (x \in \text{VALUES} \wedge x = (\ell, v) \wedge I(x))$
- $IR(x_0, x) \triangleq \exists \ell, v, v_0. (\ell \in \text{LOCATIONS} \wedge v \in \text{MEMORY} \wedge x = (\ell, v) \wedge x_0 = (\ell_0, v_0) \wedge RP_\ell(v_0, v))$
- $RP_\ell(v_0, v) \triangleq \exists x, x_0. (x, x_0 \in \text{VALUES} \wedge x = (\ell, v) \wedge x_0 = (\ell_0, v_0) \wedge IR(x_0, x))$

The transformation is based on the transition relation defined for each pair of consecutive control labels, which is expressed very simply by the following relational form :

$$x \ r_{\ell, \ell'} \ x' \triangleq (pc = \ell \wedge \text{cond}_{\ell, \ell'}(v) \wedge v' = f_{\ell, \ell'}(v) \wedge pc' = \ell')$$

Transition from one label to another label

- The transition from ℓ to ℓ' is possible when the condition $cond_{\ell,\ell'}(v)$ is true for V and when control is in ℓ ($pc = \ell$).
- When the transition is observed, the variables V are transformed as follows : $v' = f_{\ell,\ell'}(v)$.
- The definition of the transition does not express any assumptions relating to an execution strategy, such as fairness, for example.
- $cond_{\ell,\ell'}(v) \wedge v' = f_{\ell,\ell'}(v)$ is an expression in which the expressions $cond_{\ell,\ell'}(v)$ and $v' = f_{\ell,\ell'}(v)$ raise questions of definition :
 - ▶ $DOM(\ell, \ell')(v) \triangleq DEF(cond_{\ell,\ell'}(v))(v) \wedge DEF(f_{\ell,\ell'}(v))$
 - ▶ $DEF(E(X))(x)$ means that the expression $E(X)$ is defined for x , the current value of X .
- Some transitions may lead to bugs :
 - ▶ $DEF(X + 1)(x) \triangleq x + 1 \in D$, where D is the coding domain of X ; for example, $D = -2^{31}..2^{31} - 1$ for a 32-bit coding scheme.
 - ▶ $DEF(T(I + 1) < V)(t, x, v) \triangleq i + 1 \in dom(t) \wedge v \in D \wedge t(i + 1) \in D$

Relationships for program statements

$$\begin{array}{l} \ell : P_{\ell}(v_0, v) \\ V := f_{\ell, \ell'}(V) \\ \ell' : P_{\ell'}(v_0, v) \end{array}$$

Traduction

- $(pc = \ell \wedge v' = f_{\ell, \ell'}(v) \wedge pc' = \ell')$
- $cond_{\ell, \ell'}(v) \triangleq TRUE$

$$\begin{array}{l} \ell_1 : P_{\ell_1}(v_0, v) \\ \textbf{WHILE } B(V) \textbf{ DO} \\ \quad \ell_2 : P_{\ell_2}(v_0, v) \\ \quad \dots \\ \quad \ell_3 : P_{\ell_3}(v_0, v) \\ \textbf{END} \\ \ell_4 : P_{\ell_4}(v_0, v) \end{array}$$

Traduction

$$\left\{ \begin{array}{l} pc = \ell_1 \wedge b(v) \wedge v' = v \wedge pc' = \ell_2 \\ pc = \ell_1 \wedge \neg b(v) \wedge v' = v \wedge pc' = \ell_4 \\ pc = \ell_3 \wedge b(v) \wedge v' = v \wedge pc' = \ell_2 \\ pc = \ell_3 \wedge \neg b(v) \wedge v' = v \wedge pc' = \ell_4 \end{array} \right.$$

Checking the contract : what the method will be

A program P *satisfies* a contract $(pre, post)$:

- P transforms a variable v starting from an initial value v_0 and producing a final value v_f : $v_0 \xrightarrow{P} v_f$
- v_0 satisfait pre : $pre(v_0)$ and v_f satisfait $post$: $post(v_0, v_f)$
- $pre(v_0) \wedge v_0 \xrightarrow{P} v_f \Rightarrow post(v_0, v_f)$

```
requires  $pre(v_0)$   
ensures  $post(v_0, v_f)$   
variables  $V$   
[  
  begin  
     $0 : P_0(v_0, v)$   
    instruction0  
    ...  
     $i : P_i(v_0, v)$   
    ...  
    instruction $f-1$   
     $f : P_f(v_0, v)$   
  end
```

- $pre(v_0) \wedge v = v_0 \Rightarrow P_0(v_0, v)$
- $pre(v_0) \wedge P_f(v_0, v) \Rightarrow post(v_0, v)$
- conditions on the transitions ℓ, ℓ' to be defined on the basis of the induction principles.

Checking a contract : a simple example

```
variables  $U, V$   
requires  $u_0, v_0 \in \mathbb{N}$   
ensures  $u_f + v_f = u_0 + v_0$   
begin  
  0 :  $u = u_0 \wedge v = v_0 \wedge u_0, v_0 \in \mathbb{N}$   
   $U := U + 2$   
  1 :  $u = u_0 + 2 \wedge v = v_0 \wedge u_0, v_0 \in \mathbb{N}$   
   $V := V - 2$   
  2 :  $u = u_0 + 2 \wedge v = v_0 - 2 \wedge u_0, v_0 \in \mathbb{N}$   
end
```

Checking to do

$$x = (pc, v) \text{ et } J(\ell_0, v_0, pc, v) \triangleq \left[\begin{array}{l} \wedge pc \in \text{LOCATIONS} \\ \wedge v \in \text{MEMORY} \\ \dots \\ \wedge pc = \ell \Rightarrow P_\ell(v_0, v) \\ \dots \end{array} \right.$$

Let $(Th(s, c), x, \text{VALUES}, \text{INIT}(x), \{r_0, \dots, r_n\})$ be a relational model for this programme. A property $A(x)$ is a safety property for P if

$\forall x_0, x \in \text{LOCATIONS} \times \text{MEMORY}. \text{Init}(x_0) \wedge x_0 \xrightarrow{\text{NEXT}} x \Rightarrow A(x)$. We

know that this property implies that there exists a state property $I(x)$ such that the three properties hold, but we apply this verification to J :

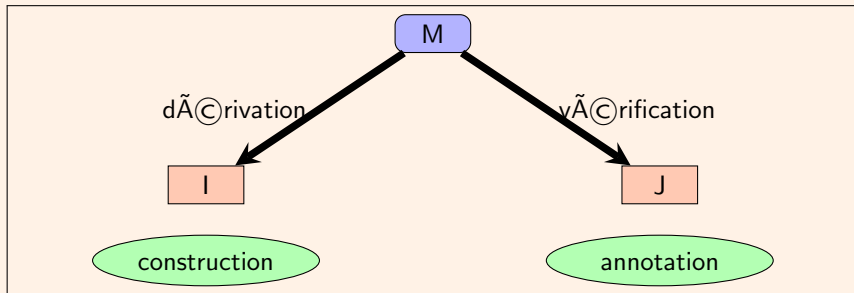
$\forall x_0, x, x' \in \text{LOCATIONS} \times \text{MEMORY}$:

$$\left\{ \begin{array}{l} (1) \quad \text{INIT}(x_0) \Rightarrow J(x_0, x_0) \\ (2) \quad J(x_0, x) \Rightarrow A(x_0, x) \\ (3) \quad \forall i \in \{0, \dots, n\} : J(x_0, x) \wedge x \ r_i \ x' \Rightarrow J(x_0, x') \end{array} \right.$$

- $\forall i \in \{0, \dots, n\} : J(x) \wedge x \ r_i \ x' \Rightarrow J(x')$ est équivalent à $J(x) \wedge (\exists i \in \{0, \dots, n\} : x \ r_i \ x') \Rightarrow J(x')$

Checking to do

- Application of the correctness to the relational induction principle :
if we verify the three properties, then A is a safety property for the model in question (verification).
- If we wish to show that A is a safety property, then we must use the invariant to construct annotations for the model (derivation).



Checking to do : back to the example

- $x = (pc, u, v)$
- $J(0, u_0, v_0, pc, u, v) \triangleq$

$$\left[\begin{array}{l} \wedge pc \in \{0, 1, 2\} \\ \wedge u, v \in \mathbb{Z} \\ \wedge pc = 0 \Rightarrow u = u_0 \wedge v = v_0 \wedge u_0, v_0 \in \mathbb{N} \\ \wedge pc = 1 \Rightarrow u = u_0 + 2 \wedge v = v_0 \wedge u_0, v_0 \in \mathbb{N} \\ \wedge pc = 2 \Rightarrow u = u_0 + 2 \wedge v = v_0 - 2 \wedge u_0, v_0 \in \mathbb{N} \end{array} \right.$$
- $A(0, u_0, v_0, pc, u, v) \triangleq (pc = 2 \Rightarrow u + v = u_0 + v_0 - 2 \wedge u_0, v_0 \in \mathbb{N})$

$$\forall pc, u, v, pc', u', v' \in \{0, 1, 2\} \times \mathbb{Z} :$$

$$\left\{ \begin{array}{l} (1) \text{ INIT}(0, u_0, v_0) \Rightarrow \text{J}(0, u_0, v_0, 0, u_0, v_0) \\ (2) \text{ J}(0, u_0, v_0, pc, u, v) \Rightarrow \text{A}(0, u_0, v_0, pc, u, v) \\ (3) \forall i \in \{0, \dots, n\} : \text{J}(0, u_0, v_0, pc, u, v) \wedge x \ r_i \ pc', u', v' \Rightarrow \text{J}(0, u_0, v_0, pc', \end{array} \right.$$

Checking the example

- 1 $\text{INIT}(0, u_0, v_0,) \Rightarrow J(0, u_0, v_0, 0, u_0, v_0) :$
 $pc = 0 \wedge u = u_0 \wedge v = v_0 \wedge u_0, v_0 \in \mathbb{N} \Rightarrow J(0, u_0, v_0, 0, u_0, v_0) :$
- 2 $J(0, u_0, v_0, pc, u, v) \Rightarrow A(0, u_0, v_0, pc, u, v)$
 $J(pc, u, v) \Rightarrow (pc = 2 \Rightarrow u + v = u_0 + v_0 - 2 \wedge u_0, v_0 \in \mathbb{N})$
- 3 $\forall i \in \{0, \dots, n\} : J(0, u_0, v_0, pc, u, v) \wedge x \ r_i \ pc', u', v' \Rightarrow$
 $J(0, u_0, v_0, pc', u', v')$
 $\left[\begin{array}{l} r01(pc, u, v, pc', u', v') \triangleq pc = 0 \wedge u' = u + 2 \wedge pc' = 1 \wedge v' = v \\ r12(pc, u, v, pc', u', v') \triangleq pc = 1 \wedge v' = v - 2 \wedge pc' = 2 \wedge u' = u \end{array} \right.$
 - ▶ $J(0, u_0, v_0, pc, u, v) \wedge r01(pc, u, v, pc', u', v') \Rightarrow J(pc', u', v')$
 - ▶ $J(0, u_0, v_0, pc, u, v) \wedge r12(pc, u, v, pc', u', v') \Rightarrow J(0, u_0, v_0, pc', u', v')$

$$J(pc, u, v) \wedge r01(pc, u, v, pc', u', v') \Rightarrow J(pc', u', v') \quad (1)$$

$$\begin{aligned} & \left(\begin{array}{l} \wedge pc \in \{0, 1, 2\} \\ \wedge u, v \in \mathbb{Z} \\ \wedge pc = 0 \Rightarrow u = u_0 \wedge v = v_0 \wedge u_0, v_0 \in \mathbb{N} \\ \wedge pc = 1 \Rightarrow u = u_0 + 2 \wedge v = v_0 \wedge u_0, v_0 \in \mathbb{N} \\ \wedge pc = 2 \Rightarrow u = u_0 + 2 \wedge v = v_0 - 2 \wedge u_0, v_0 \in \mathbb{N} \end{array} \right) \wedge \left(\begin{array}{l} \wedge pc = 0 \\ \wedge u' = u + 2 \\ \wedge pc' = 1 \\ \wedge v' = v \end{array} \right) \\ \Rightarrow & \left(\begin{array}{l} \wedge pc' \in \{0, 1, 2\} \\ \wedge u', v' \in \mathbb{Z} \\ \wedge pc' = 0 \Rightarrow u' = u_0 \wedge v' = v_0 \wedge u_0, v_0 \in \mathbb{N} \\ \wedge pc' = 1 \Rightarrow u' = u_0 + 2 \wedge v' = v_0 \wedge u_0, v_0 \in \mathbb{N} \\ \wedge pc' = 2 \Rightarrow u' = u_0 + 2 \wedge v' = v_0 - 2 \wedge u_0, v_0 \in \mathbb{N} \end{array} \right) \end{aligned}$$

Using the Rodin platform for checking those elements

The relational model $M(P)$ for the annotated program P is therefore defined as follows :

$$M(P) \triangleq (Th(s, c), (pc, v), \text{LOCATIONS} \times \text{MEMORY}, Init(\ell, v), \{r_{\ell, \ell'} \mid \ell, \ell' \in \text{LOCATIONS} \wedge \ell \longrightarrow \ell'\}).$$

The definition of $Init(x)$ depends on the precondition of P :

$$Init(x) \triangleq .x = (\ell_0, v) \wedge \mathbf{pre}(P)(v).$$

Initial conditions

The following two properties are equivalent :

- $\forall x_0 \in \text{VALUES} : Init(x_0) \Rightarrow J(x_0, x_0)$
- $\forall v \in \text{MEMORY}. \mathbf{pre}(P)(v) \wedge v = v_0 \Rightarrow P_{\ell_0}(v_0, v)$

Induction step

- Les relations r_i correspondent aux transitions satisfaisant $\ell \longrightarrow \ell'$ et on associe à chaque r_i la relation $r_{\ell, \ell'}$
 - $x \ r_{\ell, \ell'} \ x' \triangleq (pc = \ell \wedge cond_{\ell, \ell'}(v) \wedge \wedge v' = f_{\ell, \ell'}(v) \wedge pc' = \ell')$
- $J(x_0, x) \triangleq \exists v_0, \ell, v. (\ell \in \text{LOCATIONS} \wedge v_0, v \in \text{MEMORY} \wedge x = (\ell, v) \wedge P_\ell(v_0, v))$
- $P_\ell(v_0, v) \triangleq \exists x_0, x. (x_0, x \in \text{VALUES} \wedge x = (\ell, v) \wedge x_0 = (\ell_0, v_0) \wedge J(x_0, x))$

Induction step

The following two properties are equivalent :

- $\forall i \in \{0, \dots, n\} : J(x_0, x) \wedge x \text{ } r_i \text{ } x' \Rightarrow J(x_0, x')$
- $\forall \ell, \ell' \in \text{LOCATIONS} : \ell \longrightarrow \ell' \Rightarrow P_\ell(v_0, v) \wedge \text{cond}_{\ell, \ell'}(v) \wedge v' = f_{\ell, \ell'}(v) \Rightarrow P_{\ell'}(v_0, v')$

Induction step (explanation)

- $J(x_0 n x) \wedge x \ r_{\ell, \ell'} \ x' \Rightarrow J(x_0, x')$

Induction step (explanation)

- $J(x_0, n x) \wedge x \text{ } r_{\ell, \ell'} \text{ } x' \Rightarrow J(x_0, x')$
- $x \text{ } r_{\ell, \ell'} \text{ } x' \triangleq (pc = \ell \wedge cond_{\ell, \ell'}(v) \wedge \wedge v' = f_{\ell, \ell'}(v) \wedge pc' = \ell')$

Induction step (explanation)

- $J(x_0 n x) \wedge x \text{ } r_{\ell, \ell'} \text{ } x' \Rightarrow J(x_0, x')$
- $x \text{ } r_{\ell, \ell'} \text{ } x' \triangleq (pc = \ell \wedge cond_{\ell, \ell'}(v) \wedge \wedge v' = f_{\ell, \ell'}(v) \wedge pc' = \ell')$
- $I(x) \triangleq \exists \ell, v. (\ell \in \text{LOCATIONS} \wedge v \in \text{MEMORY} \wedge x = (\ell, v) \wedge P_{\ell}(v))$

Induction step (explanation)

- $J(x_0, x) \wedge x \text{ r}_{\ell, \ell'} x' \Rightarrow J(x_0, x')$
- $x \text{ r}_{\ell, \ell'} x' \triangleq (pc = \ell \wedge cond_{\ell, \ell'}(v) \wedge \wedge v' = f_{\ell, \ell'}(v) \wedge pc' = \ell')$
- $I(x) \triangleq \exists \ell, v. (\ell \in \text{LOCATIONS} \wedge v \in \text{MEMORY} \wedge x = (\ell, v) \wedge P_\ell(v))$
- $P_\ell(v_0, v) \triangleq \exists x. (x \in \text{VALUES} \wedge x = (\ell, v) \wedge x_0 = (\ell_0, v_0) \wedge J(x_0, x))$

Induction step (explanation)

- $J(x_0nx) \wedge x \text{ } r_{\ell,\ell'} \text{ } x' \Rightarrow J(x_0, x')$
- $x \text{ } r_{\ell,\ell'} \text{ } x' \triangleq (pc = \ell \wedge cond_{\ell,\ell'}(v) \wedge \wedge v' = f_{\ell,\ell'}(v) \wedge pc' = \ell')$
- $I(x) \triangleq \exists \ell, v. (\ell \in \text{LOCATIONS} \wedge v \in \text{MEMORY} \wedge x = (\ell, v) \wedge P_\ell(v))$
- $P_\ell(v_0, v) \triangleq \exists x. (x \in \text{VALUES} \wedge x = (\ell, v) \wedge x_0 = (\ell_0, v_0) \wedge J(x_0, x))$
- $J(x_0nx) \equiv pc = \ell \wedge P_\ell(v_0, v)$

Induction step (explanation)

- $J(x_0 n x) \wedge x \text{ } r_{\ell, \ell'} \text{ } x' \Rightarrow J(x_0, x')$
- $x \text{ } r_{\ell, \ell'} \text{ } x' \triangleq (pc = \ell \wedge cond_{\ell, \ell'}(v) \wedge \wedge v' = f_{\ell, \ell'}(v) \wedge pc' = \ell')$
- $I(x) \triangleq \exists \ell, v. (\ell \in \text{LOCATIONS} \wedge v \in \text{MEMORY} \wedge x = (\ell, v) \wedge P_\ell(v))$
- $P_\ell(v_0, v) \triangleq \exists x. (x \in \text{VALUES} \wedge x = (\ell, v) \wedge x_0 = (\ell_0, v_0) \wedge J(x_0, x))$
- $J(x_0 n x) \equiv pc = \ell \wedge P_\ell(v_0, v)$
- $J(x_0, x') \equiv pc = \ell' \wedge P_{\ell'}(v_0, v')$

Induction step (explanation)

- $J(x_0nx) \wedge x \text{ } r_{\ell,\ell'} \text{ } x' \Rightarrow J(x_0, x')$
- $x \text{ } r_{\ell,\ell'} \text{ } x' \triangleq (pc = \ell \wedge cond_{\ell,\ell'}(v) \wedge \wedge v' = f_{\ell,\ell'}(v) \wedge pc' = \ell')$
- $I(x) \triangleq \exists \ell, v. (\ell \in \text{LOCATIONS} \wedge v \in \text{MEMORY} \wedge x = (\ell, v) \wedge P_\ell(v))$
- $P_\ell(v_0, v) \triangleq \exists x. (x \in \text{VALUES} \wedge x = (\ell, v) \wedge x_0 = (\ell_0, v_0) \wedge J(x_0, x))$
- $J(x_0nx) \equiv pc = \ell \wedge P_\ell(v_0, v)$
- $J(x_0, x') \equiv pc = \ell' \wedge P_{\ell'}(v_0, v')$
- $pc = \ell \wedge P_\ell(v_0, v) \wedge (pc = \ell \wedge cond_{\ell,\ell'}(v) \wedge \wedge v' = f_{\ell,\ell'}(v) \wedge pc' = \ell' \Rightarrow pc = \ell' \wedge P_{\ell'}(v_0, v'))$

Induction step (explanation)

- $J(x_0 n x) \wedge x \text{ } r_{\ell, \ell'} \text{ } x' \Rightarrow J(x_0, x')$
- $x \text{ } r_{\ell, \ell'} \text{ } x' \triangleq (pc = \ell \wedge cond_{\ell, \ell'}(v) \wedge \wedge v' = f_{\ell, \ell'}(v) \wedge pc' = \ell')$
- $I(x) \triangleq \exists \ell, v. (\ell \in \text{LOCATIONS} \wedge v \in \text{MEMORY} \wedge x = (\ell, v) \wedge P_\ell(v))$
- $P_\ell(v_0, v) \triangleq \exists x. (x \in \text{VALUES} \wedge x = (\ell, v) \wedge x_0 = (\ell_0, v_0) \wedge J(x_0, x))$
- $J(x_0 n x) \equiv pc = \ell \wedge P_\ell(v_0, v)$
- $J(x_0, x') \equiv pc = \ell' \wedge P_{\ell'}(v_0, v')$
- $pc = \ell \wedge P_\ell(v_0, v) \wedge (pc = \ell \wedge cond_{\ell, \ell'}(v) \wedge \wedge v' = f_{\ell, \ell'}(v) \wedge pc' = \ell' \Rightarrow pc = \ell' \wedge P_{\ell'}(v_0, v'))$
- $pc = \ell \wedge P_\ell(v_0, v) \wedge (pc = \ell \wedge cond_{\ell, \ell'}(v) \wedge \wedge v' = f_{\ell, \ell'}(v) \wedge pc' = \ell' \Rightarrow pc = \ell') \text{ (Tautologie)}$

Induction step (explanation)

- $J(x_0 n x) \wedge x \text{ r}_{\ell, \ell'} x' \Rightarrow J(x_0, x')$
- $x \text{ r}_{\ell, \ell'} x' \triangleq (pc = \ell \wedge cond_{\ell, \ell'}(v) \wedge \wedge v' = f_{\ell, \ell'}(v) \wedge pc' = \ell')$
- $I(x) \triangleq \exists \ell, v. (\ell \in \text{LOCATIONS} \wedge v \in \text{MEMORY} \wedge x = (\ell, v) \wedge P_\ell(v))$
- $P_\ell(v_0, v) \triangleq \exists x. (x \in \text{VALUES} \wedge x = (\ell, v) \wedge x_0 = (\ell_0, v_0) \wedge J(x_0, x))$
- $J(x_0 n x) \equiv pc = \ell \wedge P_\ell(v_0, v)$
- $J(x_0, x') \equiv pc = \ell' \wedge P_{\ell'}(v_0, v')$
- $pc = \ell \wedge P_\ell(v_0, v) \wedge (pc = \ell \wedge cond_{\ell, \ell'}(v) \wedge \wedge v' = f_{\ell, \ell'}(v) \wedge pc' = \ell') \Rightarrow pc = \ell' \wedge P_{\ell'}(v_0, v')$
- $pc = \ell \wedge P_\ell(v_0, v) \wedge (pc = \ell \wedge cond_{\ell, \ell'}(v) \wedge \wedge v' = f_{\ell, \ell'}(v) \wedge pc' = \ell') \Rightarrow pc = \ell' \text{ (Tautologie)}$
- $pc = \ell \wedge P_\ell(v) \wedge (pc = \ell \wedge cond_{\ell, \ell'}(v) \wedge \wedge v' = f_{\ell, \ell'}(v) \wedge pc' = \ell') \Rightarrow P_{\ell'}(v_0, v')$

Induction step (explanation)

- $J(x_0 n x) \wedge x \text{ } r_{\ell, \ell'} \text{ } x' \Rightarrow J(x_0, x')$
- $x \text{ } r_{\ell, \ell'} \text{ } x' \triangleq (pc = \ell \wedge cond_{\ell, \ell'}(v) \wedge \wedge v' = f_{\ell, \ell'}(v) \wedge pc' = \ell')$
- $I(x) \triangleq \exists \ell, v. (\ell \in \text{LOCATIONS} \wedge v \in \text{MEMORY} \wedge x = (\ell, v) \wedge P_\ell(v))$
- $P_\ell(v_0, v) \triangleq \exists x. (x \in \text{VALUES} \wedge x = (\ell, v) \wedge x_0 = (\ell_0, v_0) \wedge J(x_0, x))$
- $J(x_0 n x) \equiv pc = \ell \wedge P_\ell(v_0, v)$
- $J(x_0, x') \equiv pc = \ell' \wedge P_{\ell'}(v_0, v')$
- $pc = \ell \wedge P_\ell(v_0, v) \wedge (pc = \ell \wedge cond_{\ell, \ell'}(v) \wedge \wedge v' = f_{\ell, \ell'}(v) \wedge pc' = \ell' \Rightarrow pc = \ell' \wedge P_{\ell'}(v_0, v'))$
- $pc = \ell \wedge P_\ell(v_0, v) \wedge (pc = \ell \wedge cond_{\ell, \ell'}(v) \wedge \wedge v' = f_{\ell, \ell'}(v) \wedge pc' = \ell' \Rightarrow pc = \ell' \text{ (Tautologie)})$
- $pc = \ell \wedge P_\ell(v) \wedge (pc = \ell \wedge cond_{\ell, \ell'}(v) \wedge \wedge v' = f_{\ell, \ell'}(v) \wedge pc' = \ell' \Rightarrow P_{\ell'}(v_0, v'))$
- $P_\ell(v_0, v) \wedge cond_{\ell, \ell'}(v) \wedge v' = f_{\ell, \ell'}(v) \Rightarrow P_{\ell'}(v_0, v')$

Conclusion

- $J(x_0, x) \triangleq \exists \ell, v, v_0. (\ell \in \text{LOCATIONS} \wedge v, v_0 \in \text{MEMORY} \wedge x = (\ell, v) \text{wedge} x_0 = (\ell_0, v_0) \wedge P_\ell(v_0, v))$
- $P_\ell(v_0, v) \triangleq \exists x. (x, x_0 \in \text{VALUES} \wedge x = (\ell, v) \text{wedge} x_0 = (\ell_0, v_0) \wedge J(x_0), x)$
- $J(x_0, x) \Rightarrow A(x_0, x)$
- $\exists \ell, v, v_0. (\ell \in \text{LOCATIONS} \wedge v, v_0 \in \text{MEMORY} \wedge x = (\ell, v) \text{wedge} x_0 = (\ell_0, v_0) \wedge P_\ell(v_0, v)) \Rightarrow A(x_0, x)$
- $\forall \ell, v, v_0. (\ell \in \text{LOCATIONS} \wedge v, v_0 \in \text{MEMORY} \wedge x = (\ell, v) \text{wedge} x_0 = (\ell_0, v_0) \wedge P_\ell(v_0, v)) \Rightarrow A(x_0, x)$
- $\forall \ell \in \text{LOCATIONS}, v, v_0 \in \text{MEMORY}. P_\ell(v_0, v) \Rightarrow A(\ell_0, v_0, \ell, v)$

Conclusion

The following two properties are equivalent :

- $J(x_0, x) \Rightarrow A(x_0, x)$
- $\forall \ell \in \text{LOCATIONS}, v, v_0 \in \text{MEMORY}. P_\ell(v_0, v) \Rightarrow A(\ell_0, v_0, \ell, v)$

The following verification conditions are equivalent :

- $\forall x_0, x, x' \in \text{LOCATIONS} \times \text{MEMORY} :$
$$\left\{ \begin{array}{l} (1) \text{ INIT}(x_0) \Rightarrow J(x_0, x_0) \\ (2) J(x_0, x) \Rightarrow A(x_0, x) \\ (3) \forall i \in \{0, \dots, n\} : J(x_0, x) \wedge x \text{ r}_i x' \Rightarrow J(x_0, x') \end{array} \right.$$
- $\forall v_0, v, v' \in \text{MEMORY} :$
$$\left\{ \begin{array}{l} (1) \text{ pre(P)}(v_0) \wedge v = v_0 \Rightarrow P_{\ell_0}(v_0, v) \\ (2) \forall \ell \in \text{LOCATIONS}. P_{\ell}(v_0, v) \Rightarrow A(\ell_0, v_0, \ell, v) \\ (3) \forall \ell, \ell' \in \text{LOCATIONS} : \\ \ell \longrightarrow \ell' \Rightarrow P_{\ell}(v_0, v) \wedge \text{cond}_{\ell, \ell'}(v) \wedge v' = f_{\ell, \ell'}(v) \Rightarrow P_{\ell'}(v_0, v') \end{array} \right.$$

A relational method for proving the correctness of safety properties

- The programme is annotated.
- The annotations define an invariant to be verified in accordance with the verification conditions.
- $A(\ell, v)$ is the statement of the safety property to be verified.

Relational method for verifying safety properties

Let $A(\ell_0, v_0, \ell, v)$ be a property of a program P . Let $\{P_\ell(v_0, v) : \ell \in \text{LOCATIONS}\}$ be a family of annotations, i.e. a family of properties, for this program. If the following conditions are satisfied :
Therefore, $A(\ell_0, v_0, \ell, v)$ is a safety property for the program P .

Definition

Condition de vérification

L'expression $P_\ell(v_0, v) \wedge \text{cond}_{\ell, \ell'}(v) \wedge v' = f_{\ell, \ell'}(v) \Rightarrow P_{\ell'}(v_0, v')$ où ℓ, ℓ' sont deux étiquettes liées par la relation $\longrightarrow$, est appelée une condition de vérification.

Floyd and Hoare

- $\forall v_0, v, v' \in \text{MEMORY}. \forall \ell, \ell' \in \text{LOCATIONS}. \ell \longrightarrow \ell' \Rightarrow$
 $P_\ell(v_0, v) \wedge \text{cond}_{\ell, \ell'}(v) \wedge v' = f_{\ell, \ell'}(v) \Rightarrow P_{\ell'}(v_0, v')$ is equivalent to
 $\forall \ell, \ell' \in \text{LOCATIONS}. \ell \longrightarrow \ell' \Rightarrow \forall v' \in$
 $\text{MEMORY}. P_\ell(v_0, v) \wedge \text{cond}_{\ell, \ell'}(v) \Rightarrow P_{\ell'}(v_0, v \mapsto f_{\ell, \ell'}(v))$
- $\forall v_0, v, v' \in \text{MEMORY}. \forall \ell, \ell' \in \text{LOCATIONS}. \ell \longrightarrow \ell' \Rightarrow$
 $P_\ell(v_0, v) \wedge \text{cond}_{\ell, \ell'}(v) \wedge v' = f_{\ell, \ell'}(v) \Rightarrow P_{\ell'}(v_0, v')$ est équivalent à
 $\forall \ell, \ell' \in \text{LOCATIONS}. \ell \longrightarrow \ell' \Rightarrow \forall v' \in$
 $\text{MEMORY}. (\exists v \in \text{MEMORY}. P_\ell(v_0, v) \wedge \text{cond}_{\ell, \ell'}(v) \wedge v' = f_{\ell, \ell'}(v)) \Rightarrow$
 $P_{\ell'}(v_0, v')$

Verification condition for the assignment

We can summarise the two possible forms of the assignment as follows :

- $\forall v, v' \in \text{MEMORY}. P_\ell(v_0, v) \wedge \text{cond}_{\ell, \ell'}(v) \wedge v' = f_{\ell, \ell'}(v) \Rightarrow P_{\ell'}(v_0, v')$

$$\begin{aligned} \ell &: P_\ell(v_0, v) \\ V &:= f_{\ell, \ell'}(V) \\ \ell' &: P_{\ell'}(v_0, v) \end{aligned}$$

Verification condition for the assignment

We can summarise the two possible forms of the assignment as follows :

- $\forall v, v' \in \text{MEMORY}. P_\ell(v_0, v) \wedge \text{cond}_{\ell, \ell'}(v) \wedge v' = f_{\ell, \ell'}(v) \Rightarrow P_{\ell'}(v_0, v')$
- $\forall v, v' \in \text{MEMORY}. P_\ell(v_0, v) \wedge \text{TRUE} \wedge v' = f_{\ell, \ell'}(v) \Rightarrow P_{\ell'}(v_0, v')$

$$\begin{aligned} \ell &: P_\ell(v_0, v) \\ V &:= f_{\ell, \ell'}(V) \\ \ell' &: P_{\ell'}(v_0, v) \end{aligned}$$

Verification condition for the assignment

We can summarise the two possible forms of the assignment as follows :

- $\forall v, v' \in \text{MEMORY}. P_\ell(v_0, v) \wedge \text{cond}_{\ell, \ell'}(v) \wedge v' = f_{\ell, \ell'}(v) \Rightarrow P_{\ell'}(v_0, v')$
- $\forall v, v' \in \text{MEMORY}. P_\ell(v_0, v) \wedge \text{TRUE} \wedge v' = f_{\ell, \ell'}(v) \Rightarrow P_{\ell'}(v_0, v')$
- $\forall v, v' \in \text{MEMORY}. P_\ell(v_0, v) \wedge v' = f_{\ell, \ell'}(v) \Rightarrow P_{\ell'}(v_0, v')$

$$\begin{array}{l} \ell : P_\ell(v_0, v) \\ V := f_{\ell, \ell'}(V) \\ \ell' : P_{\ell'}(v_0, v) \end{array}$$

Verification condition for the assignment

We can summarise the two possible forms of the assignment as follows :

- $\forall v, v' \in \text{MEMORY}. P_\ell(v_0, v) \wedge \text{cond}_{\ell, \ell'}(v) \wedge v' = f_{\ell, \ell'}(v) \Rightarrow P_{\ell'}(v_0, v')$
- $\forall v, v' \in \text{MEMORY}. P_\ell(v_0, v) \wedge \text{TRUE} \wedge v' = f_{\ell, \ell'}(v) \Rightarrow P_{\ell'}(v_0, v')$
- $\forall v, v' \in \text{MEMORY}. P_\ell(v_0, v) \wedge v' = f_{\ell, \ell'}(v) \Rightarrow P_{\ell'}(v_0, v')$
- $\forall v \in \text{MEMORY}. P_\ell(v_0, v) \Rightarrow P_{\ell'}(v_0, v \mapsto f_{\ell, \ell'}(v))$
(Hoare axiomatic system).

$$\begin{array}{l} \ell : P_\ell(v_0, v) \\ V := f_{\ell, \ell'}(V) \\ \ell' : P_{\ell'}(v_0, v) \end{array}$$

Verification condition for the assignment

We can summarise the two possible forms of the assignment as follows :

- $\forall v, v' \in \text{MEMORY}. P_\ell(v_0, v) \wedge \text{cond}_{\ell, \ell'}(v) \wedge v' = f_{\ell, \ell'}(v) \Rightarrow P_{\ell'}(v_0, v')$
- $\forall v, v' \in \text{MEMORY}. P_\ell(v_0, v) \wedge \text{TRUE} \wedge v' = f_{\ell, \ell'}(v) \Rightarrow P_{\ell'}(v_0, v')$
- $\forall v, v' \in \text{MEMORY}. P_\ell(v_0, v) \wedge v' = f_{\ell, \ell'}(v) \Rightarrow P_{\ell'}(v_0, v')$
- $\forall v \in \text{MEMORY}. P_\ell(v_0, v) \Rightarrow P_{\ell'}(v_0, v \mapsto f_{\ell, \ell'}(v))$
(Hoare axiomatic system).
- $\forall v \in \text{MEMORY}. (\exists v' \in \text{MEMORY}. P_\ell(v_0, v) \wedge v' = f_{\ell, \ell'}(v)) \Rightarrow P_{\ell'}(v_0, v')$
corresponds to Floyd assignment rule.

$$\begin{array}{l} \ell : P_\ell(v_0, v) \\ V := f_{\ell, \ell'}(V) \\ \ell' : P_{\ell'}(v_0, v) \end{array}$$

Verification condition for the iteration

```

 $\ell_1 : P_{\ell_1}(v_0, v)$ 
WHILE  $B(v)$  DO
   $\ell_2 : P_{\ell_2}(v_0, v)$ 
  ...
   $\ell_3 : P_{\ell_3}(v_0, v)$ 
END
 $\ell_4 : P_{\ell_4}(v_0, v)$ 

```

For the iteration structure, the verification conditions are as follows :

- $P_{\ell_1}(v_0, v) \wedge B(v) \Rightarrow P_{\ell_2}(v_0, v)$
- $P_{\ell_1}(v_0, v) \wedge \neg B(v) \Rightarrow P_{\ell_4}(v_0, v)$
- $P_{\ell_3}(v_0, v) \wedge B(v) \Rightarrow P_{\ell_2}(v_0, v)$
- $P_{\ell_3}(v_0, v) \wedge \neg B(v) \Rightarrow P_{\ell_4}(v_0, v)$

Verification conditions for the conditional

$$\begin{array}{l} \ell_1 : P_{\ell_1}(v_0, v) \\ \text{IF } B(v) \text{ THEN} \\ \quad \ell_2 : P_{\ell_2}(v_0, v) \\ \quad \dots \\ \quad \ell_3 : P_{\ell_3}(v_0, v) \\ \text{ELSE} \\ \quad m_2 : P_{\ell_2}(v_0, v) \\ \quad \dots \\ \quad m_3 : P_{\ell_3}(v_0, v) \\ \text{FI} \\ \ell_4 : P_{\ell_4}(v_0, v) \end{array}$$

For the conditional statement, the verification concludes

- $P_{\ell_1}(v_0, v) \wedge B(v) \Rightarrow P_{\ell_2}(v_0, v)$
- $P_{\ell_3}(v_0, v) \Rightarrow P_{\ell_4}(v_0, v)$
- $P_{\ell_1}(v_0, v) \wedge \neg B(v) \Rightarrow P_{m_2}(v_0, v)$
- $P_{m_3}(v_0, v) \Rightarrow P_{\ell_4}(v_0, v)$

Program correctness properties

Let v be a state variable of P . **pre**(P)(v) is the precondition of P for v ; it characterises the initial values of v . **post**(P)(v_0, v) is the postcondition of P for v ; it characterises the final values of v in relation to the initial value v_0

Example

- 1 **pre**(P)(x, y, z)= $x, y, z \in \mathbb{N}$ et
post(P)(x_0, y_0, z_0, x, y, z)= $z = x_0 * y_0$
- 2 **pre**(Q)(x, y, z)= $x, y, z \in \mathbb{N}$ et
post(Q)(x_0, y_0, z_0, x, y, z)= $z = x_0 + y_0$

$$\forall \underline{x}, \underline{y}, \underline{r}, \underline{q}, \overline{x}, \overline{y}, \overline{r}, \overline{q}.$$

$$\mathbf{pre}(P)(\underline{x}, \underline{y}, \underline{r}, \underline{q}) \wedge (\underline{x}, \underline{y}, \underline{r}, \underline{q}) \xrightarrow{P} (\overline{x}, \overline{y}, \overline{r}, \overline{q}) \\ \Rightarrow \mathbf{post}(P)(\underline{x}, \underline{y}, \underline{r}, \underline{q}, \overline{x}, \overline{y}, \overline{r}, \overline{q})$$

Partial Correctness of a program

Partial correctness aims to establish that a programme P is partially correct with respect to its precondition and its postcondition.

- the data specification for P **pre**(P)(v_0)
- the specification of the outputs of P **post**(P)(v_0, v)
- a family of property annotations $\{P_\ell(v_0, v) : \ell \in \text{LOCATIONS}\}$ for this programme.
- a safety property defining the partial correctness
 $pc = \ell_f \Rightarrow \mathbf{post}(P)(v_0, v_f)$, where ℓ_f is the label marking the end of the programme P

Definition

The programme P is partially correct with respect to **pre**(P)(v_0) and **post**(P)(v_0, v), if the property $pc = \ell_f \Rightarrow \mathbf{post}(P)(v_0, v)$ is a safety property for this programme.

Partial Correctness of a program

If the following conditions are met :

- $\forall v_0, v \in \text{MEMORY} : \mathbf{pre}(P)(v_0) \wedge v = v_0 \Rightarrow P_{\ell_0}(v_0, v)$
- $\forall v_0, v \in \text{MEMORY} : P_{\ell_f}(v_0, v) \Rightarrow \mathbf{post}(P)(v_0, v)$
- $\forall \ell, \ell' \in \text{LOCATIONS} : \ell \longrightarrow \ell' : \forall v_0, v, v' \in \text{MEMORY}. (P_{\ell}(v_0, v) \wedge \mathit{cond}_{\ell, \ell'}(v) \wedge v' = f_{\ell, \ell'}(v) \Rightarrow P_{\ell'}(v_0, v'))$,

So the program P is partially correct with respect to $\mathbf{pre}(P)(v_0)$ et $\mathbf{post}(P)(v_0, v)$.

- The partial correctness states that if the program terminates normally, then the postcondition is satisfied by the current variables.
- The semantics of the contract are therefore fairly straightforward to define :

Reformulation of computation

- $\text{pre}(v_0) \wedge v_0 \xrightarrow{P} v_f \Rightarrow \text{post}(v_0, v_f)$
(expression of the partial correctness)
- $pc_0 = \ell_0 \wedge \text{pre}(v_0) \wedge (pc_0, v_0) \xrightarrow{\text{NEXT}^*} (pc, v) \wedge pc = \ell_f \Rightarrow \text{post}(v_0, v_f)$
(big-step semantics and small-step semantics equivalence)
- $pc_0 = \ell_0 \wedge \text{pre}(v_0) \wedge (pc_0, v_0) \xrightarrow{\text{NEXT}^*} (pc, v) \Rightarrow (pc = \ell_f \Rightarrow \text{post}(v_0, v_f))$
(implication and conjunction property)

- $\text{Init}(x_0) \wedge x_0 \xrightarrow{\text{NEXT}^*} x \Rightarrow \text{PC}(x_0, x)$
$$\begin{aligned} \text{Init}(x_0) &\triangleq pc_0 = \ell_0 \wedge \text{pre}(v_0) \\ x_0 &\triangleq (\ell_0, v_0) \text{ and } x \triangleq (pc, v) \\ \text{PC}(x_0, x) &\triangleq x_0 = (\ell_0, v_0) \wedge x = (pc, v) \Rightarrow (pc = \ell_f \Rightarrow \text{post}(v_0, v_f)) \end{aligned}$$



Partial correctness is a safety property and the relational method for safety properties is applied.

Checking the contract

A program P *satisfies* a contract $(\text{pre}, \text{post})$:

- P transforms a variable v from an initial value v_0 to a final value v_f :
$$v_0 \xrightarrow{P} v_f$$
- v_0 satisfies pre : $\text{pre}(v_0)$ and v_f satisfies post : $\text{post}(v_0, v_f)$
- $\text{pre}(v_0) \wedge v_0 \xrightarrow{P} v_f \Rightarrow \text{post}(v_0, v_f)$

```
requires  $\text{pre}(v_0)$ 
ensures  $\text{post}(v_0, v_f)$ 
variables  $V$ 
[
  begin
     $0 : P_0(v_0, v)$ 
    statement0
    ...
     $i : P_i(v_0, v)$ 
    ...
    statement $f-1$ 
     $f : P_f(v_0, v)$ 
  end
```

- $\text{pre}(v_0) \wedge v = v_0 \Rightarrow P_0(v_0, v)$
- $\text{pre}(v_0) \wedge P_f(v_0, v) \Rightarrow \text{post}(v_0, v)$
- For any pair of labels ℓ, ℓ'
such that $\ell \longrightarrow \ell'$, we verify that,
for all values $v, v' \in \text{MEMORY}$
$$\left(\begin{array}{c} P_\ell(v_0, v) \\ \wedge \text{cond}_{\ell, \ell'}(v) \wedge v' = f_{\ell, \ell'}(v) \end{array} \right) \Rightarrow P_{\ell'}(v_0, v')$$

- The transition to be executed is the one going from ℓ to ℓ' , characterised by the condition or guard $cond_{\ell,\ell'}(v)$ on v and a transformation of the variable v , $v' = f_{\ell,\ell'}(v)$.
- An RTE condition is defined by $\mathbf{DOM}(\ell, \ell')(v)$ for the transition in question. $\mathbf{DOM}(\ell, \ell')(v)$ means that the transition $\ell \rightarrow \ell'$ is possible and does not lead to an error.
- An error is an arithmetic overflow, a reference to an array element that does not exist, a reference to a null pointer, ...

example

- 1 The transition corresponds to an assignment of the form $x := x + y$ or $y := x + y$:
$$\mathbf{DOM}(x + y)(x, y) \triangleq \mathbf{DOM}(x)(x, y) \wedge \mathbf{DOM}(y)(x, y) \wedge x + y \in \text{int}$$
- 2 The transition corresponds to an assignment of the form $x := x + 1$ ou $y := x + 1$:
$$\mathbf{DOM}(x + 1)(x, y) \triangleq \mathbf{DOM}(x)(x, y) \wedge x + 2 \in \text{int}$$

Definition RTE

The absence of run time errors is intended to establish that a program P will not produce errors during its execution with regard to its precondition and postcondition.

- the data specification for P **pre**(P)(v)
- the result specification for P **post**(P)(v_0, v)
- a family of property annotations $\{P_\ell(v) : \ell \in \text{LOCATIONS}\}$ for this programme.
- a safety property defining the absence of errors at execution :

$$\bigwedge_{\ell \in \text{LOCATIONS} - \{\text{output}\}, n \in \text{LOCATIONS}, \ell \longrightarrow n} (\mathbf{DOM}(\ell, n)(v))$$

Definition

The program P will not produce any errors during execution with respect to **pre**(P)(v) and **post**(P)(v_0, v), if the property

$\bigwedge_{\ell \in \text{LOCATIONS} - \{\text{output}\}, n \in \text{LOCATIONS}, \ell \longrightarrow n} (\mathbf{DOM}(\ell, n)(v))$ is a safety property for this program.

RTE = Run Time Error

If the following conditions are satisfied :

- $\forall v_0, v \in \text{MEMORY} : \mathbf{pre}(\mathbf{P})(v_0) \wedge v = v_0 \Rightarrow P_{\ell_0}(v_0, v)$
- $\forall m \in \text{LOCATIONS} - \{\ell_f\}, n \in \text{LOCATIONS}, \forall v_0, v, v' \in \text{MEMORY} : m \longrightarrow n : \mathbf{pre}(\mathbf{P})(v_0) \wedge P_m(v_0, v) \Rightarrow \mathbf{DOM}(m, n)(v)$
- $\forall \ell, \ell' \in \text{LOCATIONS} : \ell \longrightarrow \ell' : \forall v_0, v, v' \in \text{MEMORY}. (P_\ell(v_0, v) \wedge \text{cond}_{\ell, \ell'}(v) \wedge v' = f_{\ell, \ell'}(v) \Rightarrow P_{\ell'}(v_0, v'))$,

Therefore, the program $\mathbf{P}$ will not produce any run-time error with respect to $\mathbf{pre}(\mathbf{P})(v_0)$ and $\mathbf{post}(\mathbf{P})(v_0, v)$.

- First, one must verify the partial correction and then reinforce the assertions of the partial correctness with domain conditions.
- One can therefore derive a contract that also incorporates the verification of RTE.

Relational verification method for RTE

A program P fulfils a contract $(pre, post)$:

- P transforms a variable v from an initial value v_0 to a final value v_f :

$$v_0 \xrightarrow{P} v_f$$
- v_0 satisfies $pre : pre(v_0)$ and v_f satisfies $post : post(v_0, v_f)$
- $pre(v_0) \wedge v_0 \xrightarrow{P} v_f \Rightarrow post(v_0, v_f)$
- $\mathbb{D}$ is the RTE domain of V

```

requires pre(v0)
ensures post(v0, vf)
variables V
[
  begin
    0 : P0(v0, v)
    instruction0
    ...
    i : Pi(v0, v)
    ...
    instructionf-1
    f : Pf(v0, v)
  end

```

- $pre(v_0) \wedge v = v_0 \Rightarrow P_0(v_0, v)$
- $pre(x_0) \wedge P_f(v_0, v) \Rightarrow post(v_0, v)$
- For any pair of labels ℓ, ℓ'
 such that $\ell \longrightarrow \ell'$, we find that, for all
 values $v, v' \in \text{MEMORY}$

$$\left(\begin{array}{c} P_\ell(v_0, v) \\ \wedge cond_{\ell, \ell'}(v) \wedge v' = f_{\ell, \ell'}(v) \\ \Rightarrow P_{\ell'}(v_0, v') \end{array} \right),$$
- $\forall m \in \text{LOCATIONS} - \{\ell_f\}, n \in \text{LOCATIONS}, \forall v_0, v, v' \in \text{MEMORY} :$
 $m \longrightarrow n :$
 $pre(v_0) \wedge P_m(v_0, v) \Rightarrow \text{DOM}(m, n)(v)$

- 1 CM3
- 2 Overview of machines, contexts and proof obligations
- 3 Proof Obligations for Contexts and Machines
 - PO thm/THM (context)
 - PO th/THM (machine)
 - PO evt/inv/INV
 - PO evt/act/FIS
- 4 Method for proving invariance properties

- 5 Verification
 - Definition of machine for checking contract
 - Soundness of the technique
- 6 Conclusion

Event B Machine for checking contract of a program

A program P *satisfies* a contract $(\text{pre}, \text{post})$ and P transforms a variable v .

Event B Machine for checking contract of a program

A program P *satisfies* a contract $(pre, post)$ and P transforms a variable v .

```
requires  $pre(v_0)$   
ensures  $post(v_0, v_f)$   
variables  $V$   
  begin  
    0 :  $P_0(v_0, v)$   
    statement0  
    ...  
    i :  $P_i(v_0, v)$   
    ...  
    statementf-1  
    f :  $P_f(v_0, v)$   
  end
```

Event B Machine for checking contract of a program

A program P *satisfies* a contract $(pre, post)$ and P transforms a variable v .

```
requires  $pre(v_0)$ 
ensures  $post(v_0, v_f)$ 
variables  $V$ 
begin
  0 :  $P_0(v_0, v)$ 
  statement0
  ...
  i :  $P_i(v_0, v)$ 
  ...
  statementf-1
  f :  $P_f(v_0, v)$ 
end
```

```
machine  $M$ 
sees  $C$ 
variables  $pc, v$ 
invariants
   $I(pc, v)$ 
theorems
   $pre(v_0) \wedge v = v_0 \Rightarrow P_0(v_0, v)$ 
   $pre(v_0) \wedge P_f(v_0, v) \Rightarrow post(v_0, v)$ 
INITIALISATION  $Init(v)$ 
...
e( $\ell, \ell'$ )
...
end
```

Event B Machine for checking contract of a program

A program P *satisfies* a contract $(pre, post)$ and P transforms a variable v .

```
requires  $pre(v_0)$ 
ensures  $post(v_0, v_f)$ 
variables  $V$ 
begin
  0 :  $P_0(v_0, v)$ 
  statement0
  ...
  i :  $P_i(v_0, v)$ 
  ...
  statementf-1
  f :  $P_f(v_0, v)$ 
end
```

```
machine  $M$ 
sees  $C$ 
variables  $pc, v$ 
invariants
   $I(pc, v)$ 
theorems
   $pre(v_0) \wedge v = v_0 \Rightarrow P_0(v_0, v)$ 
   $pre(v_0) \wedge P_f(v_0, v) \Rightarrow post(v_0, v)$ 
INITIALISATION  $Init(v)$ 
...
e( $\ell, \ell'$ )
...
end
```

What are the invariants, the theorems and the events, such that M is checked if, and only if, the program P fulfils a contract $(pre, post)$

- 1 CM3
- 2 Overview of machines, contexts and proof obligations
- 3 Proof Obligations for Contexts and Machines
- 4 Method for proving invariance properties
- 5 Verification
 - Definition of machine for checking contract
 - Soundness of the technique
- 6 Conclusion

Sketch of the construction

- Writing a context C for defining constants as v_0 and properties (axioms) as $pre(v_0)$.
- Defining additional constants and predicates for expressing annotations as for instance the domain D of the problem.
- Defining invariants from annotations
- Defining theorems from conditions for checking partial correctness.
- Defining events from the graph of labels.

Defining the context

- A simple algorithm decreases a variable x initially set to a value x_0 of $\mathbb{N}$
- A set L of labels is used for annotating the algorithm.
- Two constants *maxi* and *mini* for expressing the domain of values x_0

```
variables  $x$   
requires  $x_0 \in \mathbb{N}$   
ensures  $x_f = 0$   
begin  
   $\ell_0 : \{x = x_0 \wedge x_0 \in \mathbb{N}\}$   
  while  $0 < x$  do  
     $\ell_1 : \{0 < x \leq x_0 \wedge x_0 \in \mathbb{N}\}$   
     $x := x - 1;$   
     $\ell_2 : \{0 \leq x \leq x_0 \wedge x_0 \in \mathbb{N}\}$   
  od;  
   $\ell_3 : \{x = 0\}$   
end
```

Defining the context

- A simple algorithm decreases a variable x initially set to a value x_0 of $\mathbb{N}$
- A set L of labels is used for annotating the algorithm.
- Two constants *maxi* and *mini* for expressing the domain of values x_0

```
variables  $x$ 
requires  $x_0 \in \mathbb{N}$ 
ensures  $x_f = 0$ 
begin
 $\ell_0 : \{x = x_0 \wedge x_0 \in \mathbb{N}\}$ 
while  $0 < x$  do
   $\ell_1 : \{0 < x \leq x_0 \wedge x_0 \in \mathbb{N}\}$ 
   $x := x - 1$ ;
   $\ell_2 : \{0 \leq x \leq x_0 \wedge x_0 \in \mathbb{N}\}$ 
od;
 $\ell_3 : \{x = 0\}$ 
end
```

```
CONTEXT CONTEXT
SETS  $L$ 
CONSTANTS  $x_0$   $l_0$   $l_1$   $l_2$   $l_3$  mini maxi
AXIOMS
  @pre  $x_0 \in \mathbb{N}$ 
  @axm2 partition( $L$ ,  $\{l_0\}$ ,  $\{l_1\}$ ,  $\{l_2\}$ ,  $\{l_3\}$ )
  @axm3 mini  $\in \mathbb{Z} \wedge$  maxi  $\in \mathbb{Z}$ 
  @axm4 mini  $< 0 \wedge$  maxi  $> 0$ 
  @axm5  $x_0 \leq$  maxi
end
```

From annotations to invariants

$$\begin{array}{l} inv_i : pc \in \text{LOCATIONS} \\ inv_j : v \in \text{Type} \\ \cdot \\ inv_k : pc = \ell \Rightarrow P_\ell(v) \\ inv_m : pc = \ell' \Rightarrow P_{\ell'}(v) \\ \cdot \\ th_n : A(pc, v) \end{array}$$

- $Type$ is the type of the variables v and is a set of possible values defined in the context C .
- The annotation is giving us for free the conditions satisfied by v when the control is in ℓ , (resp. in ℓ').
- $A(c, v)$ is a safety property that we are supposed to check and the case of Event-B, it is a theorem.

Defining the invariant for our example

- The counter variable is ofte denoted pc or l or c .
- Theorems are stating the conditions of initialistion and finalisation and the RTE conditions.

INVARIANTS

$@inv1\ x \in \mathbb{Z}$

$@inv2\ l \in L$

$@inv3\ l = @xtruc1$

$l0 \Rightarrow x = x0 \wedge x0 \in \mathbb{N}$

$@inv4\ l = l1 \Rightarrow 0 < x \wedge x \leq x0 \wedge x0 \in \mathbb{N}$

$@inv5\ l = l2 \Rightarrow 0 \leq x \wedge x \leq x0 \wedge x0 \in \mathbb{N}$

$@inv6\ l = l3 \Rightarrow x = 0$

$theorem\ @precond\ x = x0 \wedge x0 \in \mathbb{N} \Rightarrow x = x0 \wedge x0 \in \mathbb{N}$

$theorem\ @postcond\ x = 0 \Rightarrow x = 0$

$theorem\ @rtel\ x \in \mathbb{Z}$

$theorem\ @overflow\ mini \leq x \wedge x \leq maxi$

Defining events for transition steps

$$\begin{array}{c} \ell : \{P_\ell(v)\} \\ \text{cond}_{\ell, \ell'}(v) \longrightarrow v := f_{\ell, \ell'}(v) \\ \ell' : \{P_{\ell'}(v)\} \end{array}$$

```

 $e(\ell, \ell')$ 
WHEN
   $pc = \ell$ 
   $cond_{\ell, \ell'}(v)$ 
THEN
   $pc := \ell'$ 
   $v := f_{\ell, \ell'}(v)$ 
END

```

$$\begin{array}{l} \ell_0^1 : \{x = 0\} \\ \quad x := x + 1; \\ \ell_0^1 : \{x = 1\} \end{array}$$

- v is the state memory variable or list of memory variables; v includes the local variables and the results variables.
- pc is a new variable which is modelling the control flow and its type is `LOCATIONS`.
- $e(\ell, \ell')$ is simulating the computation flow starting from ℓ and moving to ℓ' ; v is updated.

Defining the events for the example

```
EVENT INITIALISATION
  then
    @act1  $x := x0$ 
    @act2  $l := l0$ 
  end
EVENT el0l1
  where
    @grd1  $l = l0$ 
    @grd2  $0 < x$ 
  then
    @act1  $l := l1$ 
  end
EVENT el0l3
  where
    @grd1  $l = l0$ 
    @grd2  $\neg(0 < x)$ 
  then
    @act1  $l := l3$ 
  end
```

```
EVENT el1l2
  where
    @grd1  $l = l1$ 
  then
    @act1  $l := l2$ 
    @act2  $x := x - 1$ 
  end
EVENT el2l1
  where
    @grd1  $l = l2$ 
    @grd2  $0 < x$ 
  then
    @act1  $l := l1$ 
  end
EVENT el2l3
  where
    @grd1  $l = l2$ 
    @grd2  $\neg(0 < x)$ 
  then
    @act1  $l := l3$ 
  end
```

- 1 CM3
- 2 Overview of machines, contexts and proof obligations
- 3 Proof Obligations for Contexts and Machines
- 4 Method for proving invariance properties
- 5 Verification
 - Definition of machine for checking contract
 - Soundness of the technique
- 6 Conclusion

Verification conditions to check for PC

- $pre(v_0) \wedge v = v_0 \Rightarrow P_0(v_0, v)$
- $pre(v_0) \wedge P_f(v_0, v) \Rightarrow post(v_0, v)$
- PC (Partial Correctness)

For any pair of labels ℓ, ℓ'

such that $\ell \longrightarrow \ell'$, We check that, for all values $v, v' \in \text{MEMORY}$

$$\left(\begin{array}{l} P_\ell(v_0, v) \\ \wedge cond_{\ell, \ell'}(v) \wedge v' = f_{\ell, \ell'}(v) \\ \Rightarrow P_{\ell'}(v_0, v') \end{array} \right),$$

i

- $pre(v_0) \wedge v = v_0 \Rightarrow P_0(v_0, v)$ **THEOREM**
- $pre(v_0) \wedge P_f(v_0, v) \Rightarrow post(v_0, v)$ **THEOREM**
- PC (Partial Correctness)

For any event $e(\ell, \ell')$ with $\ell \longrightarrow \ell'$,

for all values $v, v' \in \text{MEMORY}$

$$\left(\begin{array}{l} P_\ell(v_0, v) \\ \wedge B A(e(\ell, \ell'))(pc, v, pc', v') \end{array} \right) \Rightarrow P_{\ell'}(v_0, v')$$

- Introduction of the definition of events between two successive labels.

$$BA(\mathbf{e}(\ell, \ell'))(pc, v, pc', v')$$

For each pair of successive labels ℓ, ℓ' , the three statements are equivalent :

- $P_\ell(v) \wedge \text{cond}_{\ell,\ell'}(v) \wedge v' = f_{\ell,\ell'}(v) \Rightarrow P_{\ell'}(v')$
- $I(pc, v) \wedge pc = \ell \wedge \text{cond}_{\ell,\ell'}(v) \wedge pc' = \ell' \wedge v' = f_{\ell,\ell'}(v) \Rightarrow (pc' = \ell' \Rightarrow P_{\ell'}(v'))$
- $I(pc, v) \wedge BA(e(\ell, \ell'))(pc, v, pc', v') \wedge pc = \ell \Rightarrow (pc' = \ell' \Rightarrow P_{\ell'}(v'))$

Partial Correctness

Let AA an annotated algorithm with precondition $\mathbf{pre}(AA)(v)$ and postcondition $\mathbf{post}(AA)(v_0, v)$. Let the context C and the machine M generated from AA using the construction given previously. We assume that ℓ_0 is the first label and ℓ_e is the last label. We add the following properties in the machine M :

- $pc = \ell_0 \wedge \mathbf{pre}(\mathbf{AA})(v) \Rightarrow P_{\ell_0}(v)$ (condition of initialisation as THEOREM)
- $pc = \ell_e \Rightarrow (P_{\ell_e}(v) \Rightarrow \mathbf{post}(\mathbf{AA})(v_0, v))$ (condition of safety as THEOREM).

If proof obligations are discharged, then the annotated algorithm AA is partially correct with respect to its pre/post specification.

- The first added theorem is expressing the sound initialisation.
- The second added property is expressing the partial correctness.
- The discharged proof obligations are corresponding to the induction steps.

A relational method for proving the correctness of safety properties

- The contract is stated for variables v namely $v : [pre(v_0, post(v_0, v_f)]$
- The program is annotated on the set of labels denoted $LOCATIONS$ ($\ell \in LOCATIONS$).
- The annotations define an invariant to be verified according to the verification conditions.
- $A(\ell, v)$ is the statement of the safety property to be verified.

A relational method for proving safety properties

Let $A(\ell_0, v_0, \ell, v)$ be a property of a program P . Let $\{P_\ell(v_0, v) : \ell \in LOCATIONS\}$ be a family of annotations, i.e. a family of properties, for this programme. If the following conditions are satisfied :

$\forall v, v' \in MEMORY :$

$$\left\{ \begin{array}{l} (1) \quad Pre(v) \Rightarrow P_{\ell_0}(v) \\ (2) \quad \forall \ell \in LOCATIONS. P_\ell(v) \Rightarrow A(\ell, v) \\ (3) \quad \forall \ell, \ell' \in LOCATIONS : \\ \quad \ell \longrightarrow \ell' \Rightarrow P_\ell(v) \wedge cond_{\ell, \ell'}(v) \wedge v' = f_{\ell, \ell'}(v) \Rightarrow P_{\ell'}(v') \end{array} \right. ,$$

Therefore, $A(\ell_0, v_0, \ell, v)$ is a safety property for the program P .

Checking using the verification Machine

- Adding the properties (2) $\forall \ell \in \text{LOCATIONS}. P_\ell(v) \Rightarrow A(\ell, v)$ as THEOREMS
- Safety properties as RTE (Run Time Error), ADF (Absence of Deadlock Freedom), ...
- Expression of safety properties using the contexte or the theory plugin facilities.

From Floyd to Hoare

From Floyd to Hoare

- $\forall x_f, x_0. \text{pre}(x_0) \wedge x_0 \xrightarrow{P} x_f \Rightarrow \text{post}(x_0, x_f)$

From Floyd to Hoare

- $\forall x_f, x_0. \mathbf{pre}(x_0) \wedge x_0 \xrightarrow{P} x_f \Rightarrow \mathbf{post}(x_0, x_f)$
- $\forall x_f, x_0. \mathbf{pre}(x_0) \Rightarrow x_0 \xrightarrow{P} x_f \Rightarrow \mathbf{post}(x_0, x_f)$

From Floyd to Hoare

- $\forall x_f, x_0. \text{pre}(x_0) \wedge x_0 \xrightarrow{P} x_f \Rightarrow \text{post}(x_0, x_f)$
- $\forall x_f, x_0. \text{pre}(x_0) \Rightarrow x_0 \xrightarrow{P} x_f \Rightarrow \text{post}(x_0, x_f)$
- $\forall x_f, x_0. \text{pre}(x_0) \Rightarrow x_0 \xrightarrow{P} x_f \Rightarrow \text{post}(x_0, x_f)$

From Floyd to Hoare

- $\forall x_f, x_0. \text{pre}(x_0) \wedge x_0 \xrightarrow{P} x_f \Rightarrow \text{post}(x_0, x_f)$
- $\forall x_f, x_0. \text{pre}(x_0) \Rightarrow x_0 \xrightarrow{P} x_f \Rightarrow \text{post}(x_0, x_f)$
- $\forall x_f, x_0. \text{pre}(x_0) \Rightarrow x_0 \xrightarrow{P} x_f \Rightarrow \text{post}(x_0, x_f)$
- $\forall x_0. \text{pre}(x_0) \Rightarrow \forall x_f. x_0 \xrightarrow{P} x_f \Rightarrow \text{post}(x_0, x_f)$

From Floyd to Hoare

- $\forall x_f, x_0. \text{pre}(x_0) \wedge x_0 \xrightarrow{P} x_f \Rightarrow \text{post}(x_0, x_f)$
- $\forall x_f, x_0. \text{pre}(x_0) \Rightarrow x_0 \xrightarrow{P} x_f \Rightarrow \text{post}(x_0, x_f)$
- $\forall x_f, x_0. \text{pre}(x_0) \Rightarrow x_0 \xrightarrow{P} x_f \Rightarrow \text{post}(x_0, x_f)$
- $\forall x_0. \text{pre}(x_0) \Rightarrow \forall x_f. x_0 \xrightarrow{P} x_f \Rightarrow \text{post}(x_0, x_f)$
- $\forall x_0. \text{pre}(x_0) \Rightarrow [P] \text{post}(x_0, x_f)$
- wlp calculus is introduced

From Floyd to Hoare

- $\forall x_f, x_0. \text{pre}(x_0) \wedge x_0 \xrightarrow{P} x_f \Rightarrow \text{post}(x_0, x_f)$
- $\forall x_f, x_0. \text{pre}(x_0) \Rightarrow x_0 \xrightarrow{P} x_f \Rightarrow \text{post}(x_0, x_f)$
- $\forall x_f, x_0. \text{pre}(x_0) \Rightarrow x_0 \xrightarrow{P} x_f \Rightarrow \text{post}(x_0, x_f)$
- $\forall x_0. \text{pre}(x_0) \Rightarrow \forall x_f. x_0 \xrightarrow{P} x_f \Rightarrow \text{post}(x_0, x_f)$
- $\forall x_0. \text{pre}(x_0) \Rightarrow [P]\text{post}(x_0, x_f)$
- wlp calculus is introduced
- $[x := e]P(x) = P[x \mapsto e]$

From Floyd to Hoare

- $\forall x_f, x_0. \text{pre}(x_0) \wedge x_0 \xrightarrow{P} x_f \Rightarrow \text{post}(x_0, x_f)$
- $\forall x_f, x_0. \text{pre}(x_0) \Rightarrow x_0 \xrightarrow{P} x_f \Rightarrow \text{post}(x_0, x_f)$
- $\forall x_f, x_0. \text{pre}(x_0) \Rightarrow x_0 \xrightarrow{P} x_f \Rightarrow \text{post}(x_0, x_f)$
- $\forall x_0. \text{pre}(x_0) \Rightarrow \forall x_f. x_0 \xrightarrow{P} x_f \Rightarrow \text{post}(x_0, x_f)$
- $\forall x_0. \text{pre}(x_0) \Rightarrow [P] \text{post}(x_0, x_f)$
- wlp calculus is introduced
- $[x := e]P(x) = P[x \mapsto e]$
- $[\text{if } b(x) \text{ then } S1 \text{ else } S2]P(x) = b(x) \wedge [S1]P(x) \vee \text{not } b(x) \wedge [S2]P(x)$

From Floyd to Hoare

- $\forall x_f, x_0. \text{pre}(x_0) \wedge x_0 \xrightarrow{P} x_f \Rightarrow \text{post}(x_0, x_f)$
- $\forall x_f, x_0. \text{pre}(x_0) \Rightarrow x_0 \xrightarrow{P} x_f \Rightarrow \text{post}(x_0, x_f)$
- $\forall x_f, x_0. \text{pre}(x_0) \Rightarrow x_0 \xrightarrow{P} x_f \Rightarrow \text{post}(x_0, x_f)$
- $\forall x_0. \text{pre}(x_0) \Rightarrow \forall x_f. x_0 \xrightarrow{P} x_f \Rightarrow \text{post}(x_0, x_f)$
- $\forall x_0. \text{pre}(x_0) \Rightarrow [P] \text{post}(x_0, x_f)$
- wlp calculus is introduced
- $[x := e]P(x) = P[x \mapsto e]$
- $[\text{if } b(x) \text{ then } S1 \text{ else } S2]P(x) = b(x) \wedge [S1]P(x) \vee \text{not } b(x) \wedge [S2]P(x)$
- Frama-c uses the HOARE logic for defining the verification conditions as R. Leino in DAFNY.

From Floyd to Hoare

- $\forall x_f, x_0. \text{pre}(x_0) \wedge x_0 \xrightarrow{P} x_f \Rightarrow \text{post}(x_0, x_f)$
- $\forall x_f, x_0. \text{pre}(x_0) \Rightarrow x_0 \xrightarrow{P} x_f \Rightarrow \text{post}(x_0, x_f)$
- $\forall x_f, x_0. \text{pre}(x_0) \Rightarrow x_0 \xrightarrow{P} x_f \Rightarrow \text{post}(x_0, x_f)$
- $\forall x_0. \text{pre}(x_0) \Rightarrow \forall x_f. x_0 \xrightarrow{P} x_f \Rightarrow \text{post}(x_0, x_f)$
- $\forall x_0. \text{pre}(x_0) \Rightarrow [P] \text{post}(x_0, x_f)$
- wlp calculus is introduced
- $[x := e]P(x) = P[x \mapsto e]$
- $[\text{if } b(x) \text{ then } S1 \text{ else } S2]P(x) = b(x) \wedge [S1]P(x) \vee \text{not } b(x) \wedge [S2]P(x)$
- Frama-c uses the HOARE logic for defining the verification conditions as R. Leino in DAFNY.
- Questions of termination require the wp calculus ...

Current Section Summary

Summary on the technique for verifying safety properties

- Formalizing safety properties as state properties
- Defining the context for expressing the precondition and the postcondition if necessary.
- Using the verification machine, as an engine for checking safety properties and especially the contract for partial correctness.
- Handling termination by using convergent events and variants.
- Version using the Foyd method contrary to Frama-c.